

Exp 8 Multirate Signal Processing

8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

Program (Matlab Code)

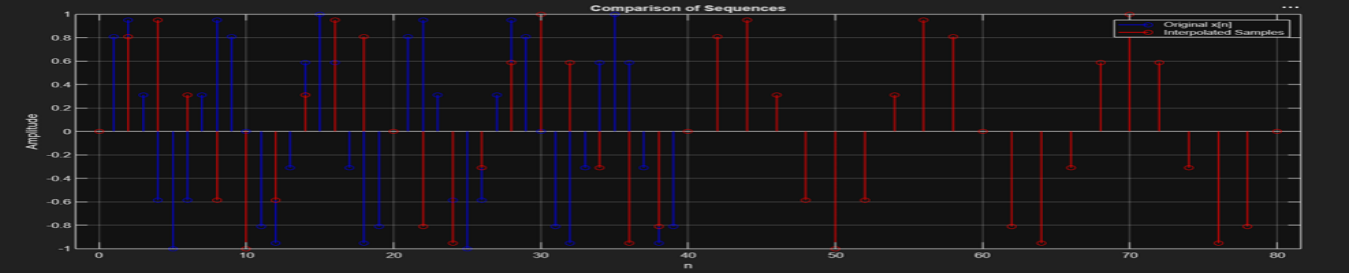
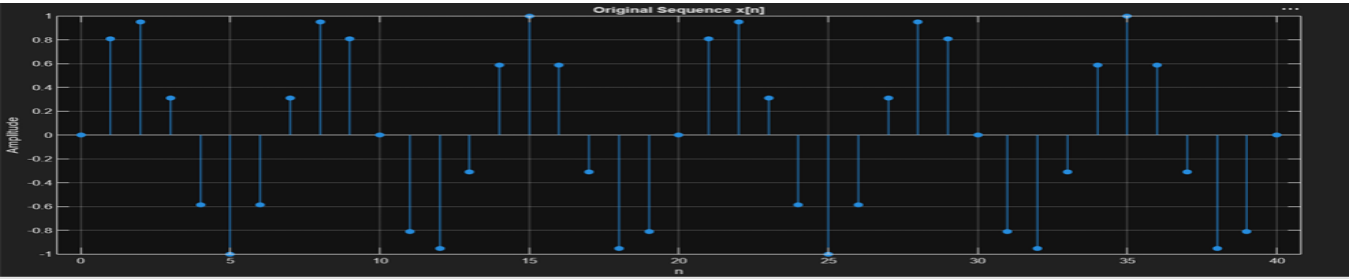
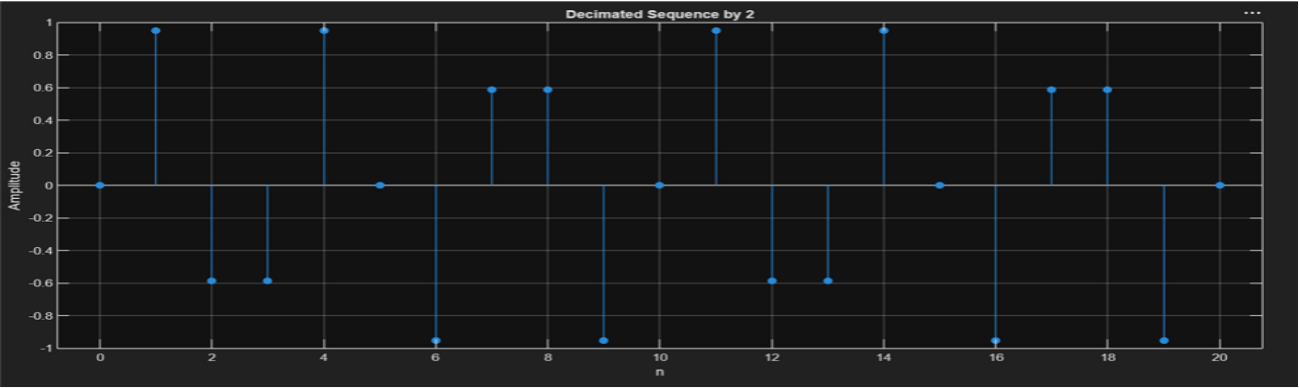
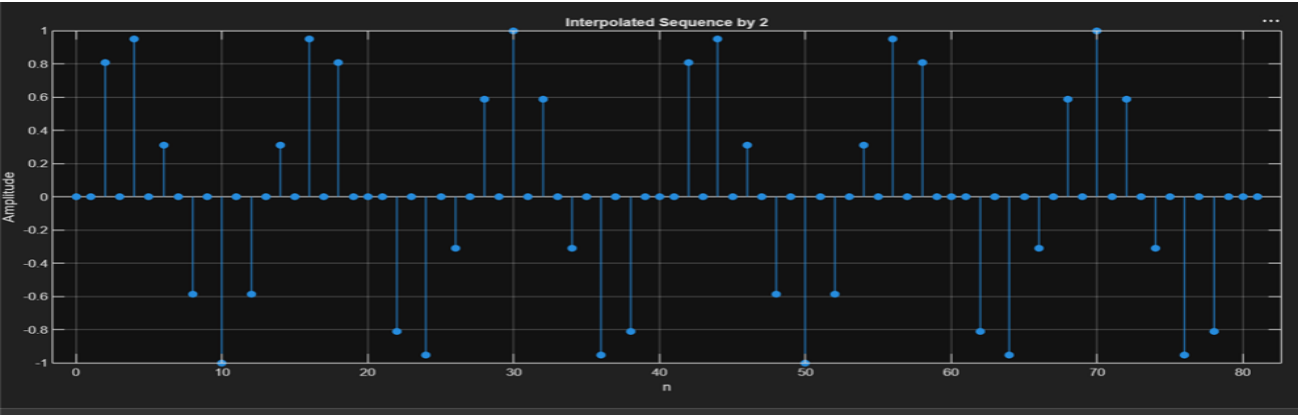
```
clc;
clear;
close all;
% Original Sequence
n = 0:40;
x = sin(0.3*pi*n);
% Decimation by 2
y = x(1:2:end);
nd = 0:length(y)-1;
% Interpolation by 2
v = zeros(1,2*length(x));
v(1:2:end) = x;
ni = 0:length(v)-1;
% Figure 1: Original Sequence
figure;
stem(n,x,'filled');
grid on;
title('Original Sequence x[n]');
```

```

xlabel('n');
ylabel('Amplitude');
% Figure 2: Decimated Sequence
figure;
stem(nd,y,'filled');
grid on;
title('Decimated Sequence by 2');
xlabel('n');
ylabel('Amplitude');
% Figure 3: Interpolated Sequence
figure;
stem(ni,v,'filled');
grid on;
title('Interpolated Sequence by 2');
xlabel('n');
ylabel('Amplitude');
% Figure 4: Comparison
figure;
stem(n,x,'b'); hold on;
stem(0:2:length(v)-2,v(1:2:end),'r');
legend('Original x[n]','Interpolated Samples');
grid on;
title('Comparison of Sequences');
xlabel('n');
ylabel('Amplitude');

```

OUTPUT:



8.2 Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB

PROGRAM

```
clc;
clear;
close all;
%% Input Signal
Fs = 1000;           % Sampling Frequency
t = 0:1/Fs:1;
% Test Signal
%Time Vector
x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t);
%% FIR Low-Pass Filter Design
N = 20;              % Filter Order
Fc = 0.4;
% Normalized Cutoff Frequency
h = fir1(N,Fc,hamming(N+1));
%% Decimation Factor
M = 2;
% Polyphase Decimation
y_dec = upfirdn(x,h,1,M);
%% Interpolation Factor
L = 2;
% Polyphase Interpolation
```

```

y_int = upfirdn(y_dec,h,L,1);
%% Adjust Length for Comparison
if length(y_int) > length(x)
    y_int = y_int(1:length(x));
elseif length(y_int) < length(x)
    x = x(1:length(y_int));
end
%% Time Axes
t1 = (0:length(x)-1)/Fs;
t2 = (0:length(y_dec)-1)/(Fs/M);
t3 = (0:length(y_int)-1)/Fs;
%% BEST GRAPH OUTPUT
figure('Name','Polyphase Decimation and
Interpolation','NumberTitle','off');
% Original Signal
subplot(4,1,1)
plot(t1,x,'b','LineWidth',1.5)
title('Original Input Signal')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Decimated Signal
subplot(4,1,2)
stem(t2,y_dec,'r','filled')
title('Decimated Signal (M = 2)')

```

```

xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Interpolated Signal
subplot(4,1,3)
plot(t3,y_int,'k','LineWidth',1.5)
title('Interpolated/Reconstructed Signal (L = 2)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Comparison
subplot(4,1,4)
plot(t1,x,'b','LineWidth',1.5)
hold on
plot(t3,y_int,'r--','LineWidth',1.5)
title('Comparison of Original and Reconstructed Signals')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on
%% Display Information
disp('-----');
disp('POLYPHASE DECIMATION & INTERPOLATION');
disp('-----');
fprintf('Original Signal Length  = %d\n',length(x));

```

```
fprintf('Decimated Signal Length    = %d\n',length(y_dec));  
fprintf('Interpolated Signal Length = %d\n',length(y_int));  
%% Separate Figure for Better Visualization  
figure;  
plot(t1,x,'b','LineWidth',2)  
hold on  
plot(t3,y_int,'r--','LineWidth',2)  
title('Original vs Reconstructed Signal')  
xlabel('Time (s)')  
ylabel('Amplitude')  
legend('Original Signal','Reconstructed Signal')  
grid on
```

OUTPUT

POLYPHASE DECIMATION & INTERPOLATION

Original Signal Length = 1001

Decimated Signal Length = 511

Interpolated Signal Length = 1001

